ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ASSIGNMENT – 5

Apply the KNN Algorithm for Classification

Dataset: Scikit-learn Diabetes Dataset

1. Aim

To implement the K-Nearest Neighbors (KNN) algorithm for classification and identify a suitable value of K for predicting the progression class in the Diabetes dataset.

2. Objectives

- Understand instance-based and distance-based classification.
- Convert the continuous target into two classes.
- Standardize features before distance calculation.
- Evaluate K values from 1 to 20.
- Measure performance using accuracy, precision, recall and F1-score.
- Visualize the confusion matrix and model performance.

3. Dataset Description

The Scikit-learn Diabetes dataset contains 442 observations and 10 independent features: age, sex, bmi, bp, s1, s2, s3, s4, s5 and s6. For regression experiments, the continuous target is used directly. For classification experiments, the continuous target was converted into two classes using the median value of 140.5: Lower Progression and Higher Progression.

Property	Value
Samples	442
Independent features	10
Classes	2
Median threshold	140.5
Best K	7
Test samples	89

4. Theory

K-Nearest Neighbors is a supervised, instance-based learning algorithm. Instead of learning an explicit parametric model during training, KNN stores the training observations and makes a prediction for a new observation based on the labels of nearby training observations.

For a query point, the algorithm calculates a distance between the query and training samples. The K nearest samples are selected, and the majority class among those neighbors is used as the predicted class for classification.

Euclidean distance is commonly used for continuous features and can be expressed as $d(x,y) = \sqrt{\Sigma(x_i-y_i)^2}$. Other distance measures can also be used depending on the type of data and problem.

The value of K controls the local neighborhood used for prediction. A very small K can make the model sensitive to noise and individual observations, while a very large K can make the model overly smooth and less sensitive to local structure.

Feature scaling is particularly important for KNN because predictions are based on distance. If one feature has a larger numerical scale than another, it can dominate the distance calculation. Standardization therefore helps ensure that the ten features contribute more comparably.

KNN has a simple training phase but can be computationally expensive during prediction because distances from the new observation to many training observations may need to be calculated.

The classification threshold in this experiment is the median target value of 140.5. This converts the regression-style target into Lower Progression and Higher Progression classes for the purpose of demonstrating classification.

Accuracy measures overall correctness, while precision measures the reliability of positive predictions. Recall measures how many actual positive cases are identified, and F1-score balances precision and recall.

The choice of K is an important model-selection decision. In this practical, values from 1 through 20 were tested and the value with the highest recorded test accuracy was selected.

KNN is often described as a lazy learner because it does not construct a conventional model during training. Most of the computational work is deferred until a prediction is requested.

The neighborhood interpretation of KNN makes the algorithm intuitive: observations that are close in feature space are assumed to have similar class labels. This makes the algorithm useful when class boundaries are irregular or nonlinear.

K should not be considered independently of the dataset size and noise level. An odd K is often used in binary classification to reduce the chance of a tie. The optimal value is normally selected using validation or cross-validation rather than relying solely on one test set.

A confusion matrix is especially useful for KNN because it reveals whether errors are concentrated in one class. This can be more informative than reporting only overall accuracy.

5. Methodology

- Load the Diabetes dataset and create binary classes using the median target.
- Split the dataset into 353 training and 89 testing observations.
- Standardize the ten independent features.
- Evaluate K values from 1 to 20.
- Select K = 7 based on the highest recorded test accuracy in the supplied results.
- Generate predictions for the test set.
- Calculate accuracy, precision, recall and F1-score.
- Generate the K-versus-accuracy graph, confusion matrix, class distribution and performance comparison graph.

6. Outputs

6.1 K Value and Accuracy

K	Accuracy
1	0.7079
2	0.7303
3	0.6742
4	0.7191
5	0.7528
6	0.7753
7	0.8090
8	0.7640
9	0.7865
10	0.7640
11	0.7978
12	0.7978
13	0.7753
14	0.7865
15	0.7753
16	0.7753
17	0.7640
18	0.7865
19	0.7528
20	0.7753

6.2 K vs Accuracy

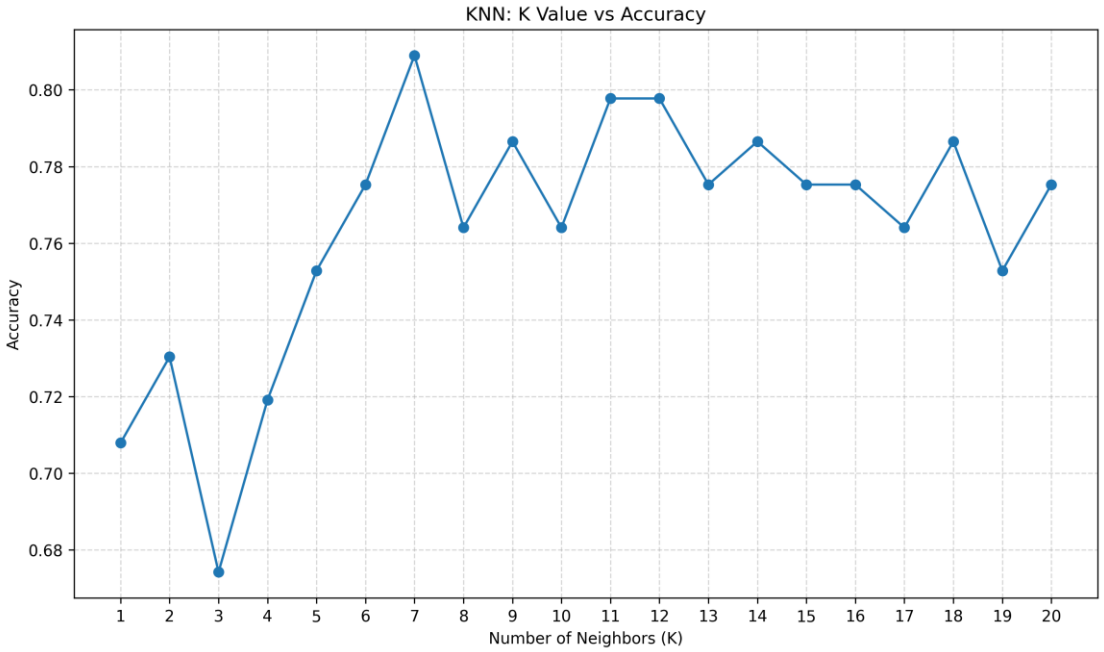


Figure 1: Accuracy for different K values

6.3 Confusion Matrix

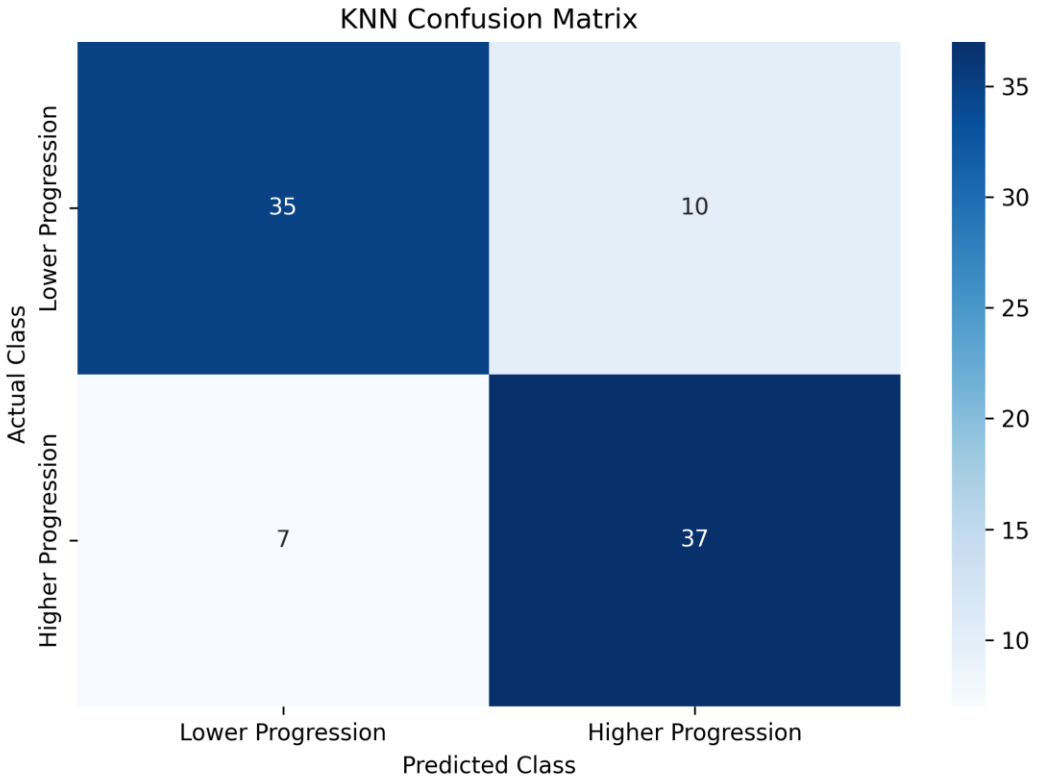


Figure 2: KNN confusion matrix

6.4 Class Distribution

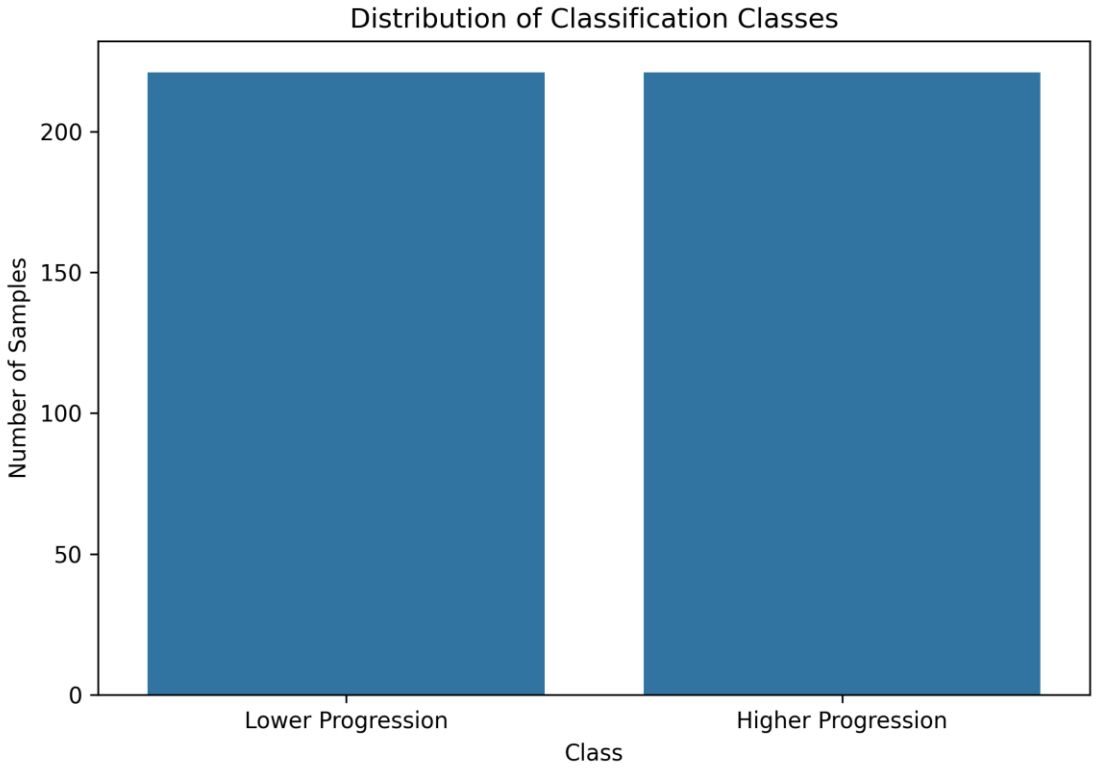


Figure 3: Class distribution

6.5 Model Performance

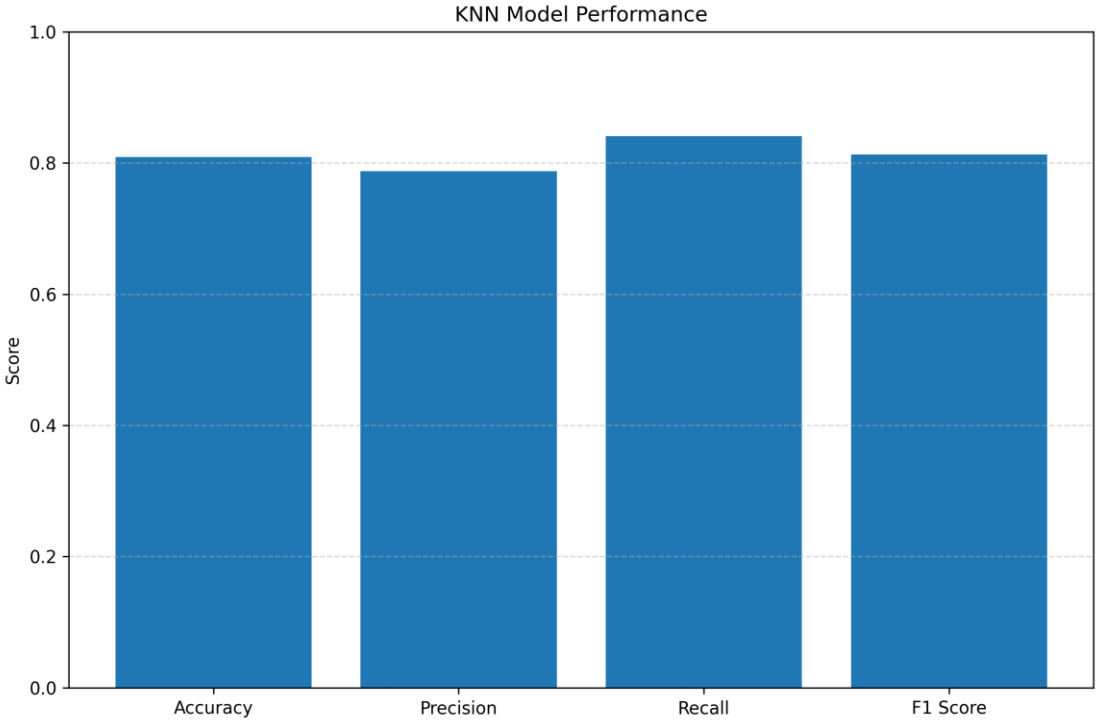


Figure 4: KNN performance metrics

7. Result

The KNN classifier achieved 80.90% accuracy, with precision of 0.7872, recall of 0.8409 and F1-score of 0.8132. K = 7 produced the highest recorded accuracy among the tested values.

8. Observations

- The ten features were standardized before distance-based classification.
- Accuracy improved to 80.90% at K = 7 among the tested values.
- The confusion matrix shows 35 correct Lower Progression predictions and 37 correct Higher Progression predictions.
- Higher Progression recall was 0.8409, indicating good identification of that class in the test set.
- KNN performance depends strongly on feature scaling and the choice of K.

9. Conclusion

K-Nearest Neighbors was successfully applied to the binary Diabetes classification problem. Testing multiple neighborhood sizes showed that K = 7 gave the best recorded test accuracy. The experiment demonstrates how distance-based classification and appropriate feature scaling can be used for practical classification tasks.